

Q1 Series Q

Concepts: what scRNA-seq sees, how it works, how to read the plots

Using a tumour as the worked example: three mental models in 30 minutes — what is in a tissue, where the data comes from, what the pipeline does.

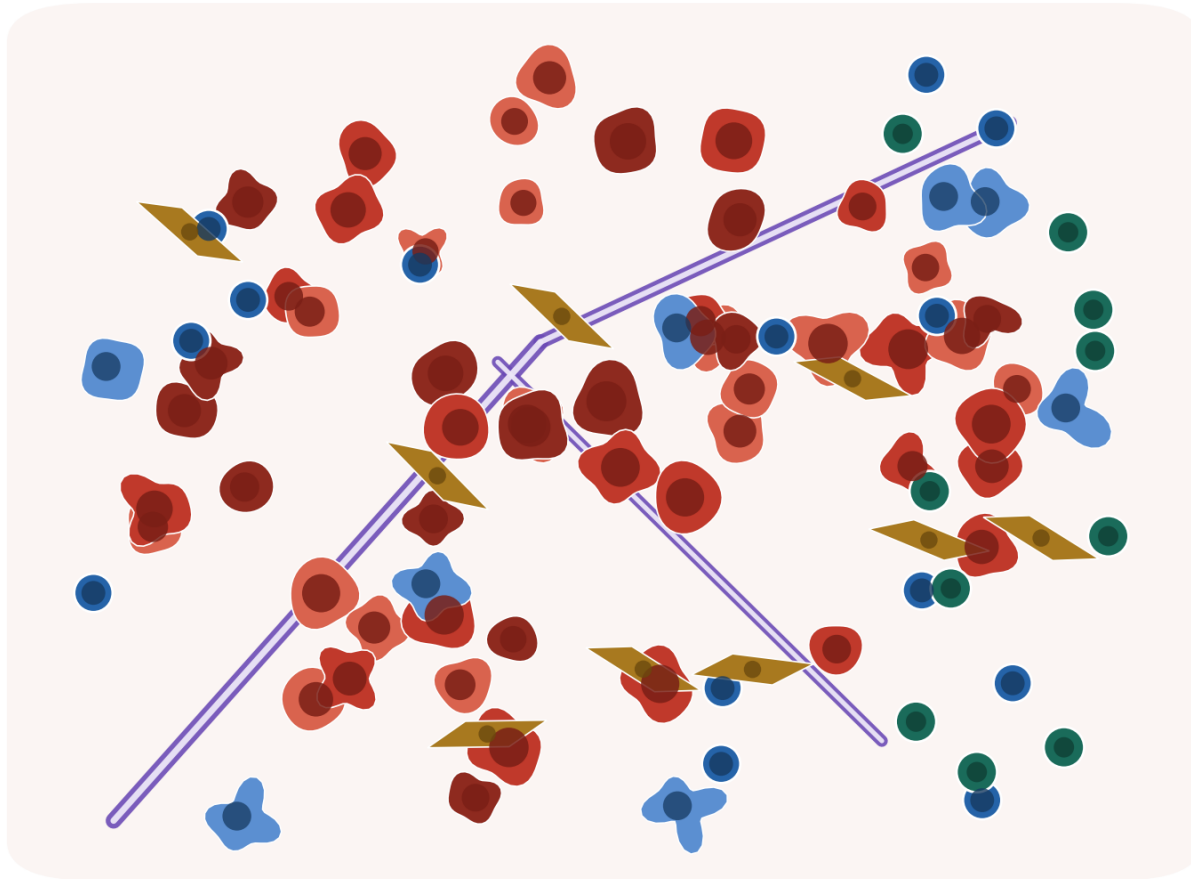
About 30 min | Prerequisites: none | Example: glioblastoma (GBM) | Deeper: Series A

A SCENE

**A 58-year-old patient, a 4 cm brain tumour on MRI.
What is actually inside the tissue the surgeon removes?**

Pathology says: glioblastoma, grade IV. That name gives you a diagnosis — not the contents of the tissue.

A tumour is an ecosystem, not a lump of malignant cells



one tumour section (schematic)

A tumour is an ecosystem, not a lump of malignant cells

	malignant cells: several states in one tumour	40–80%
	immune: macrophages /microglia, T cells	10–40%
	stroma: fibroblasts, pericytes	5–15%
	vascular endothelium	1–5%
	invaded normal tissue (GBM: oligodendrocytes, neurons)	5–30%

Proportions vary hugely by cancer, site and sampling—exactly what bulk cannot read

Malignant cells are only part of it, and come in several states; the rest is immune, stroma, vessels and invaded normal brain

WHERE THIS EPISODE STARTS

**Treatment failure, relapse, resistance —
the answer is usually 'which cell, in which state'.**

Answering that needs a technology that looks one cell at a time. This episode explains why, and what the data looks like.

In 30 minutes you will be able to...

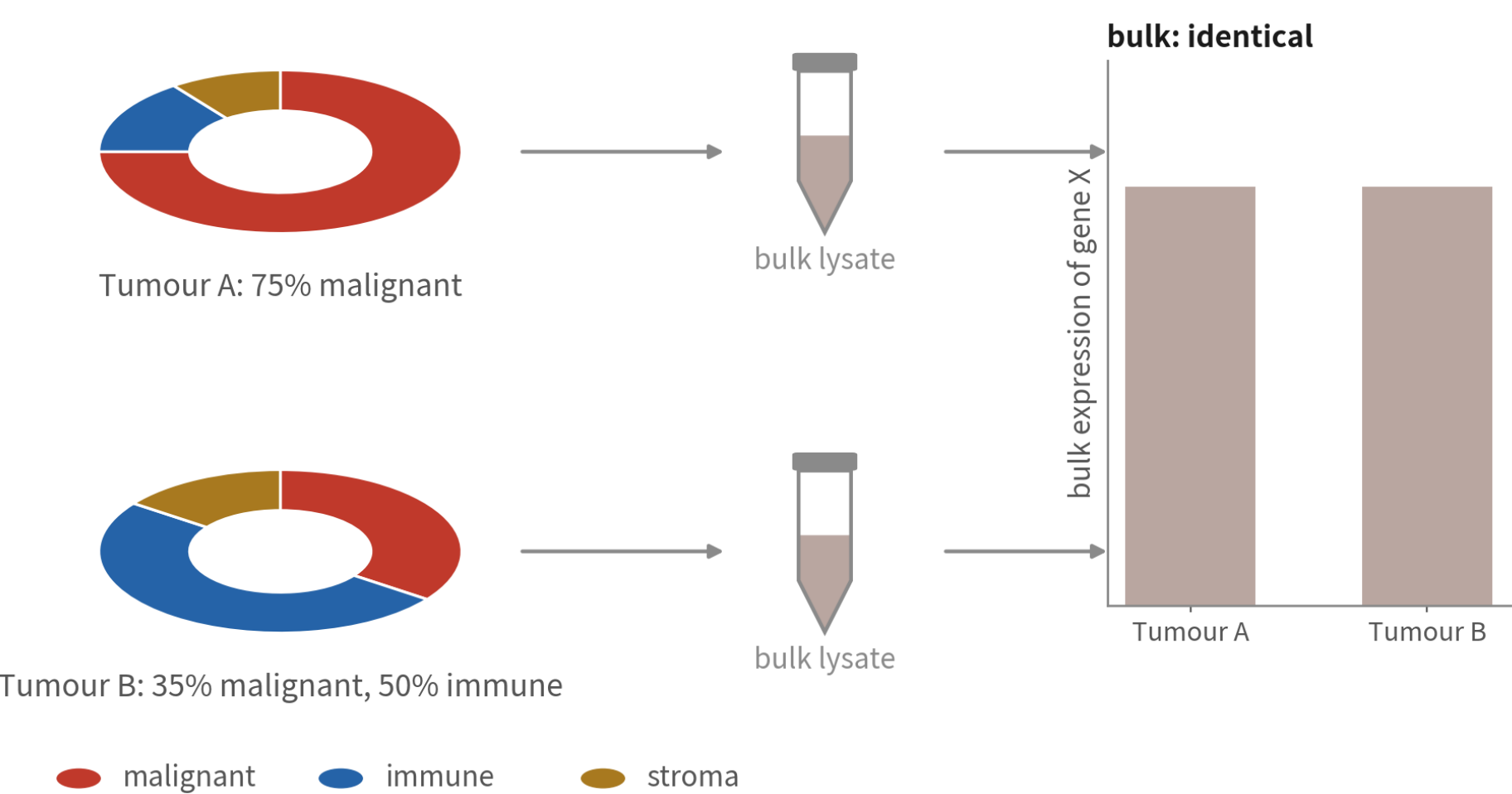
- 1 Name three tumour questions bulk cannot answer, and say why n is patients, not cells
- 2 Explain how 10x turns a tube of cells into a matrix, and what else sneaks into it
- 3 Draw the nine steps, name the two tumour-specific phenomena (malignant cells cluster by patient; CNV calls malignancy), and read a UMAP correctly

01

What the traditional method sees, and misses

Understand bulk's limits first; single cell's value then becomes obvious

Bulk RNA-seq: grind the whole tissue, measure one average



**Same bulk number,
two completely
different tumours**

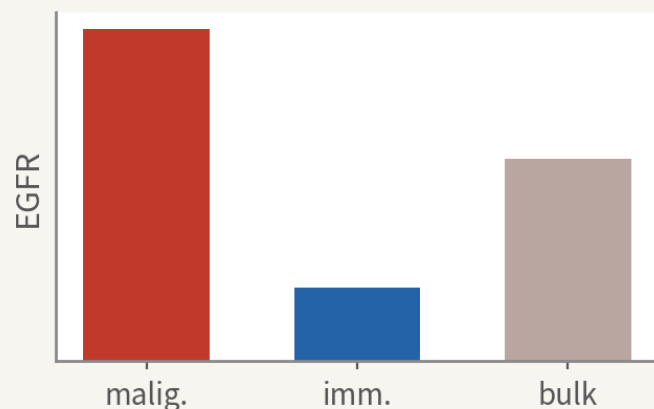
Gene X is high in malignant cells and in immune cells—summed in bulk, you cannot tell who produced the signal, or in what proportion

In oncology this is the 'purity' problem: every bulk profile is diluted by stroma and immune cells, by an unknown factor

Two tumours with completely different composition can give identical bulk numbers — oncology calls this the purity problem

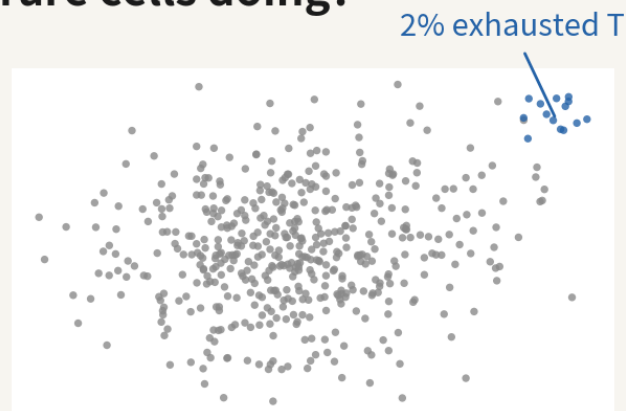
Three tumour questions bulk cannot answer

① Who produced the signal?



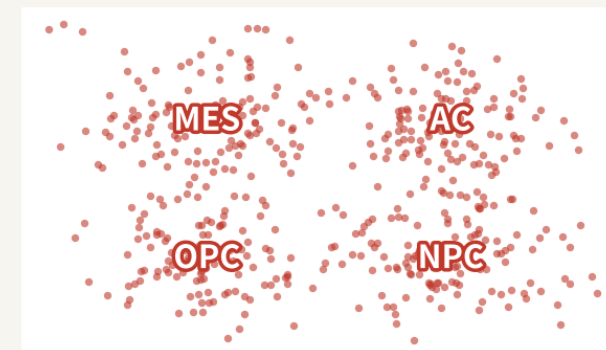
EGFR rises in the tumour—did malignant cells change, or did their proportion change? Bulk cannot tell; single cell sees each cell

② What are the rare cells doing?



Exhausted T cells or cancer stem cells may be 1–2%. Invisible in a bulk average; in single cell they form their own cluster

③ How many states do malignant cells have?

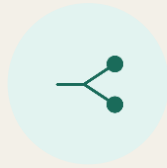


GBM malignant cells occupy four states (MES/AC/OPC/NPC-like) and switch between them. Bulk gives one average

Who produced the signal, what the rare cells are doing, how many states malignant cells occupy — each one clinically real

One phenomenon, two names

Composition change



Immune cells were recruited;
proportions shifted, each
malignant cell unchanged

Intrinsic change



Proportions unchanged, but
each malignant cell's
programme shifted

Bulk cannot separate them



Both read as 'gene X up' in bulk;
the therapeutic implications are
opposite

The most important sentence: n is patients, not cells

Design 1: one patient, 30,000 cells



30,000

n = 1

statistical unit

Design 2: six patients, 5,000 each



5,000



5,000



5,000



5,000



5,000



5,000

n = 6

supports patient-level
inference

30,000 cells from one patient share genome, age, drugs and
dissociation—they are not independent observations

30,000 cells from one patient is still n = 1. Decide how many patients at design time — Q3 shows what happens otherwise

When you actually do not need single cell

Single cell is expensive, noisy and low-powered — pin the question down before choosing the tool

'Did the whole change?'



Overall change after treatment, prognostic subtyping, cohorts of hundreds — bulk is cheaper, steadier and better powered

'Did proportions shift?' with known types



Flow cytometry, IHC or bulk deconvolution (see Q3) answer this; single cell would be an expensive counter

'Who, in which state, talking to whom?'



Rare populations, unknown types, malignant states, cell–cell communication — these are the questions only single cell can answer

02

From a piece of tissue to a matrix

Three words are enough: GEM, barcode, UMI

Operating table to sequencer: four steps



Dissociation

Enzymes break tissue into a single-cell suspension. Tumours are tough and necrotic; this step is brutal and leaves marks



Partitioning

A microfluidic chip seals each cell with a bead in an oil droplet (GEM)



Library

Each cell's mRNA is tagged with its bead's barcode; everything is pooled into one library



Sequencing & counting

Cell Ranger sorts reads back to cells by barcode, de-duplicates by UMI, builds the matrix

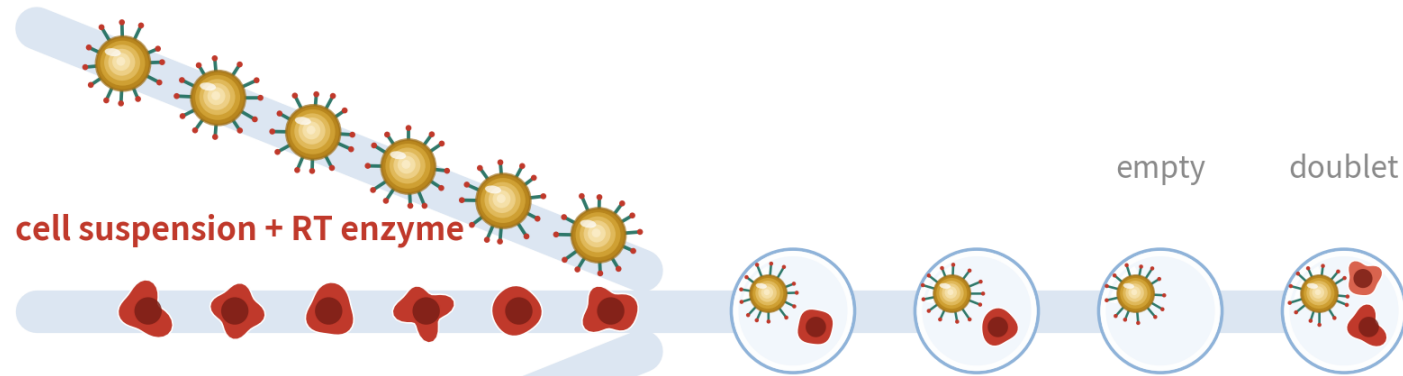
More than one platform: 3', 5', full-length, nuclei

This series uses two: 10x 3' in Q2, Smart-seq2 in Q3. The differences set QC and CNV parameters

Platform	What is read	Cells / depth	Best for	Watch out
10x 3' (Q2)	3' end of mRNA + UMI	5–10k cells 20–50k reads	Composition, types, states the default	Doublers, ambient RNA
10x 5' + V(D)J	5' end + TCR / BCR sequences	as above	Immunotherapy, T / B clonotypes	No extra information on tumour cells
Smart-seq2 (Q3)	Full-length transcripts, no UMI	Hundreds–thousands ~1M reads	CNV, splicing, mutations few precious cells	Costly, slow reads ≠ molecules
snRNA-seq (nuclei)	Nuclear RNA, mostly unspliced	as 10x	Frozen tissue, brain dissociation-fragile tissue	mt ≈ 0 immune cells under- represented

A GEM: one cell, one bead, one droplet

gel beads (millions of barcoded primers each)

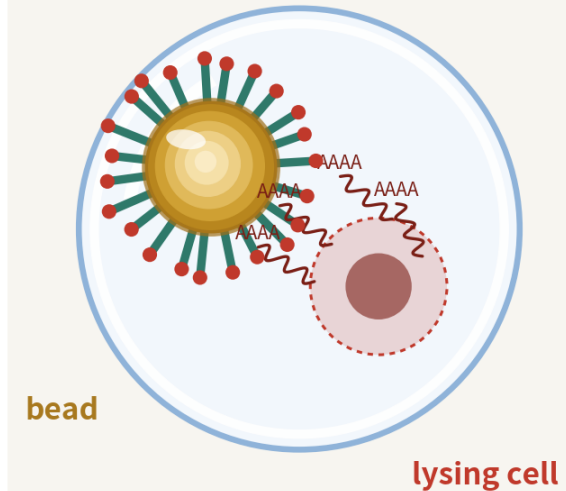


Oil pinches the aqueous stream into droplets: GEMs

oil

Bead-cell pairing is random:
most droplets are empty, a few
catch two cells (doublets)

Inside one GEM



**1 cell+1 bead+
lysis buffer**

The cell lyses beside the
bead; every mRNA's poly(A)
tail is captured on that
bead's primers—the bead's
barcode becomes the cell's
ID

Bead-cell pairing is random: most droplets are empty, a few catch two cells. Both become QC's job later

The barcode records which cell; the UMI records which molecule

Every primer on a bead



Same bead: identical barcodes → which cell



Every primer a different UMI → which molecule

So the matrix holds molecule counts (UMIs), not reads
—far smaller than bulk read counts, and more honest

The numbers in the matrix are molecules, not reads — the most fundamental difference from bulk

Why UMIs

Three identical reads:

AGTCCA

AGTCCA

AGTCCA

same UMI→PCR
copies of one
molecule, count 1

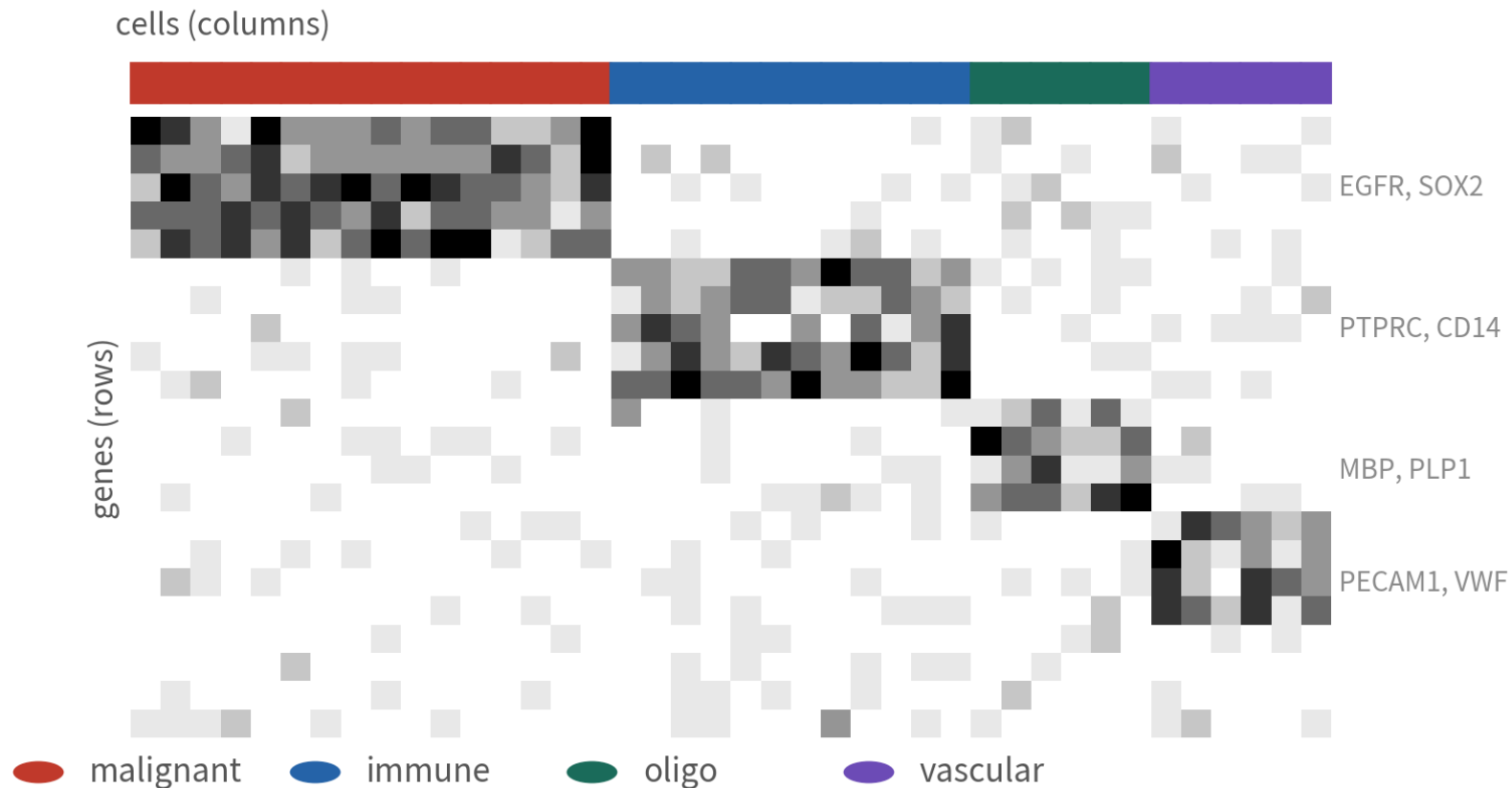
AGTCCA

TTGACG

different UMI→two
molecules, count 2

What you end up with

What Cell Ranger hands you: a gene $\times$ cell count matrix



GBM 5k example: $\sim 36\text{k}$ genes $\times$ 5,604 cells; each entry is a UMI count, $>90\%$ of entries are zero

One column per cell, one row per gene; over 90% zeros — the nature of the technology, not broken data

Three files

barcodes.tsv

one barcode per column

features.tsv

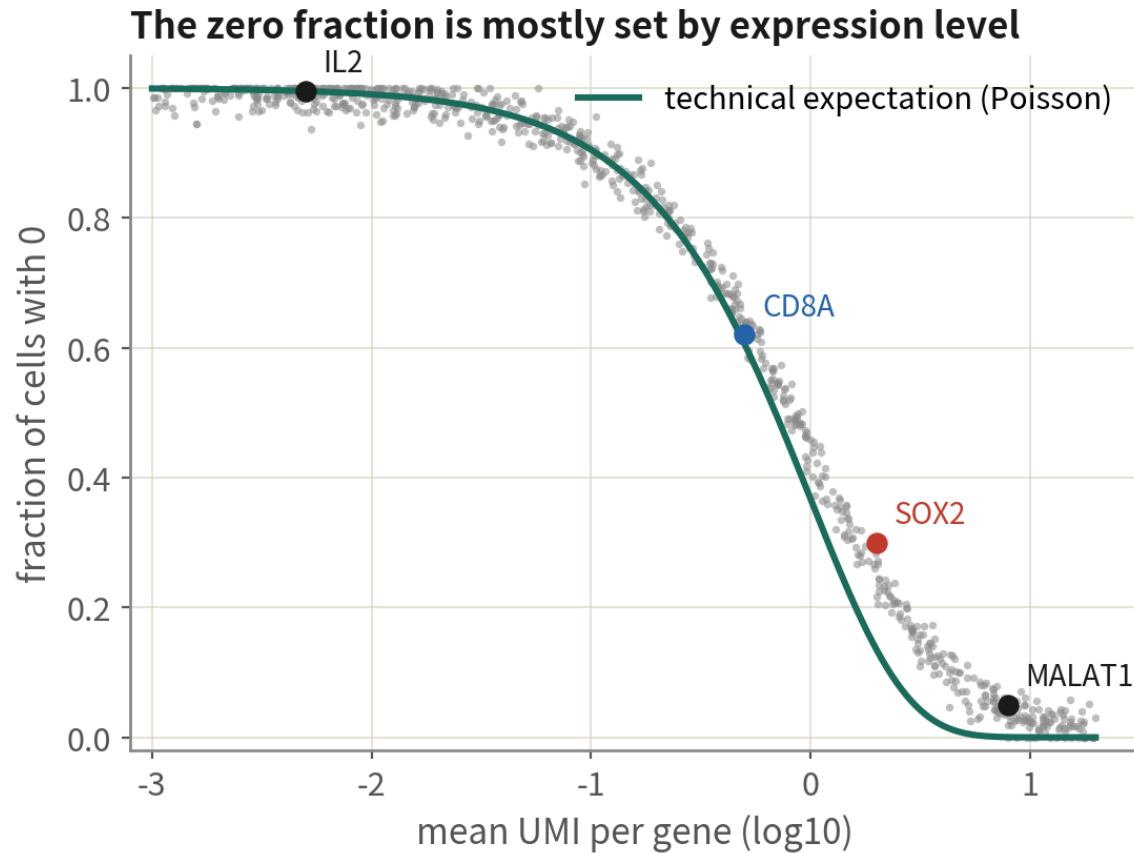
one gene per row

matrix.mtx

only non-zero
entries (sparse)

Ninety percent zeros: two kinds of zero, and conclusions rest on groups of cells

Ninety percent zeros: the nature of the technology, and the starting point of analysis



A zero has two possible meanings

- 1 truly not expressed**
e.g. SOX2 in a T cell—this zero is signal, and it is what we want
- 2 expressed but not captured (dropout)**
Only 10–30% of a cell's mRNA is captured; low genes are often missed. This zero is noise—recovered by looking at cells as a group
So: never conclude from one gene in one cell; conclusions rest on the distribution across a group of cells

The zero fraction is mostly set by expression level — technical expectation, not broken data. One gene in one cell means nothing; that is why every analysis looks at groups

Not every column is a cell



dying/broken cells

Tumour dissociation is brutal: necrotic regions, enzymatic digestion; percent.mt spikes and is often bimodal



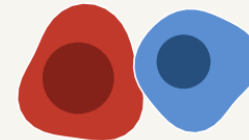
empty droplets (ambient RNA)

Necrosis releases lots of free RNA; empties carry malignant genes and look like low-quality malignant cells



doublets

A malignant cell stuck to a macrophage reads as a 'tumour cell expressing immune genes'—it has happened in the literature



dissociation stress

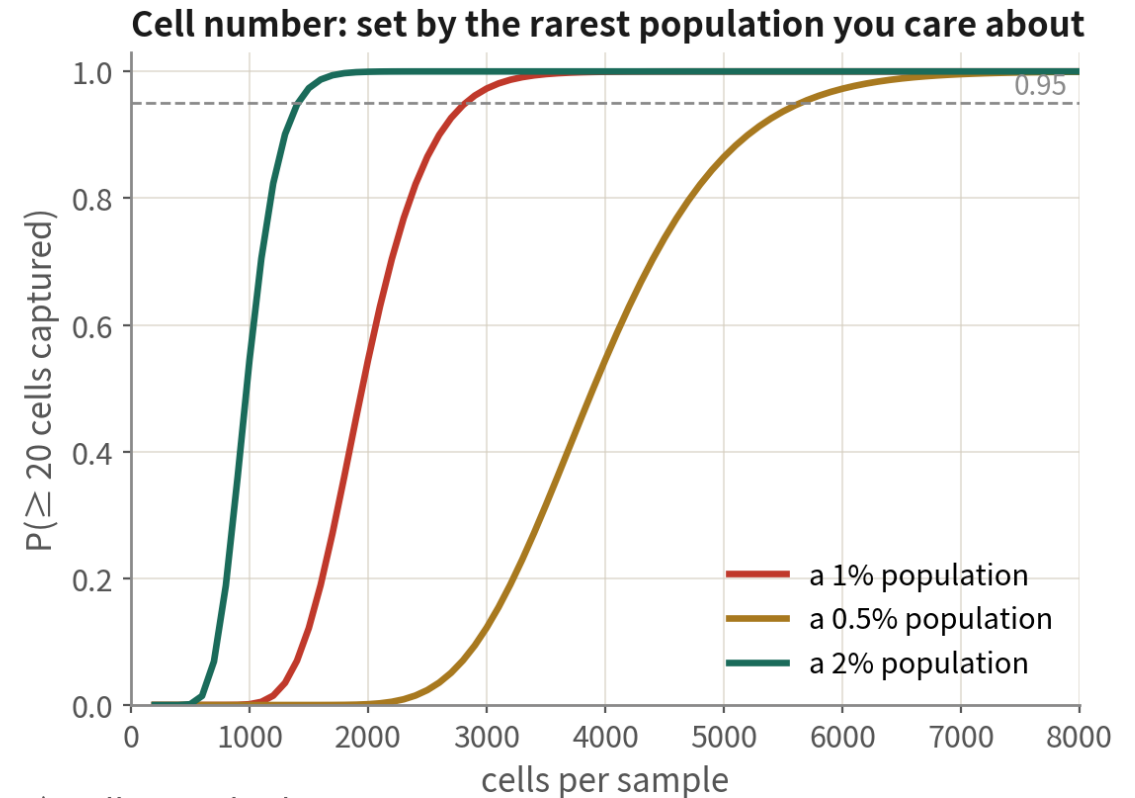
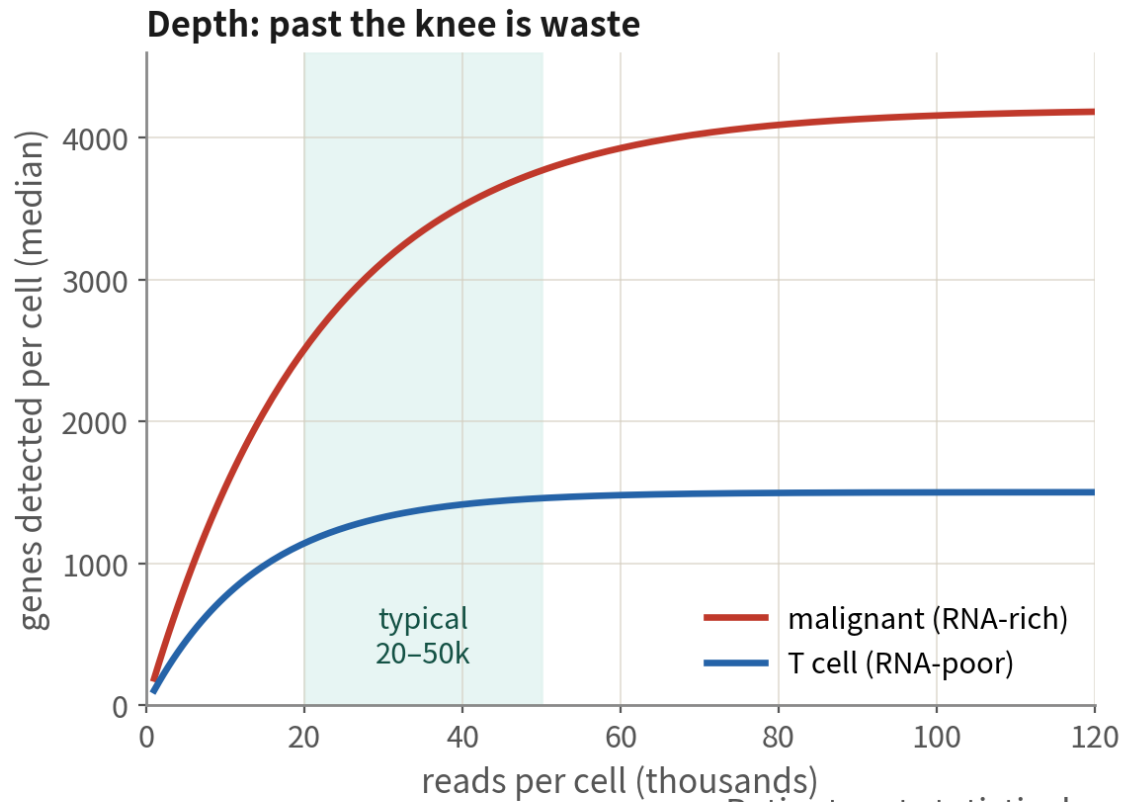
FOS/JUN/HSPA1A-high cells scattered across types. Not junk, but their 'activation' is an artefact



Tumour samples have plenty of all four. Each has a barcode and looks like a cell — hence QC is step one

Three numbers in study design: patients, cells, depth

Three numbers in study design: patients, cells per sample, reads per cell



Patients set statistical power (Q3); cells set whether rare populations are visible; past the knee, spend on more patients instead

Decide in this order: patients (power), then cells (rarest population), then depth (waste past the knee). On a fixed budget, one more patient nearly always beats 10,000 more cells

03

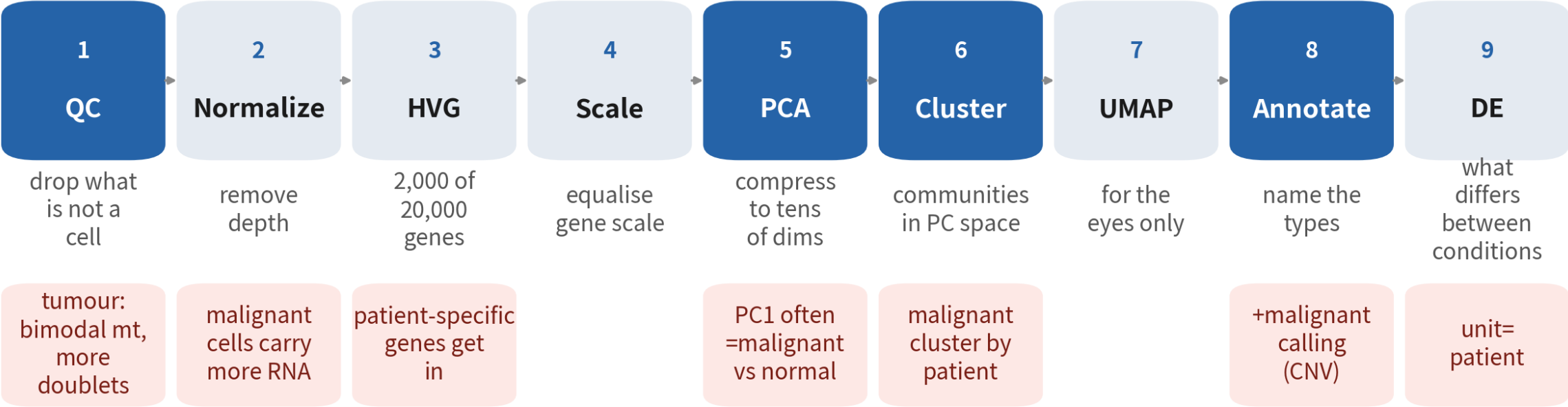
What the pipeline does

Remember the problem each step solves and you have the pipeline

Nine steps, one problem each

The nine standard steps: one problem each

Dark = steps that matter most for tumour data; red cards = what is special about tumours



The standard pipeline ends at DE. Every road after it (malignant calling, communication, trajectory, deconvolution) needs the right data—Q3 gives you the table

This is the route Q2 runs line by line on a real GBM sample. Red cards mark what is special about tumours

Analysis happens in PC space; the UMAP is a projector

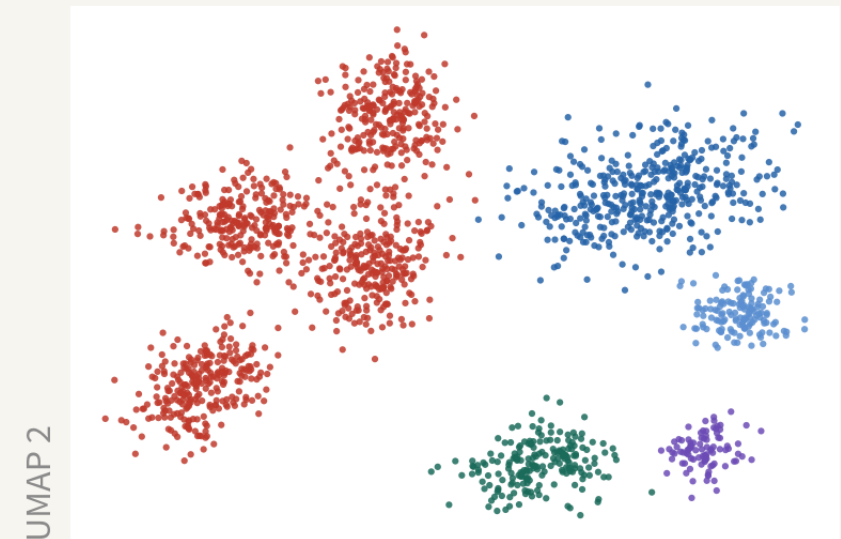
PC space (10–30 dims, invisible)

- clustering: KNN graph+communities
- cell–cell distances, neighbours
- integration (Harmony corrects here)
- trajectory, pseudotime

RunUMAP

projection

UMAP (2 dims, visible)



For looking only: which cells resemble each other. Distance, area and shape are not evidence

Clustering, distances, integration and trajectories all happen on the left; the right is just for your eyes

04

What is special about the example: two tumour phenomena

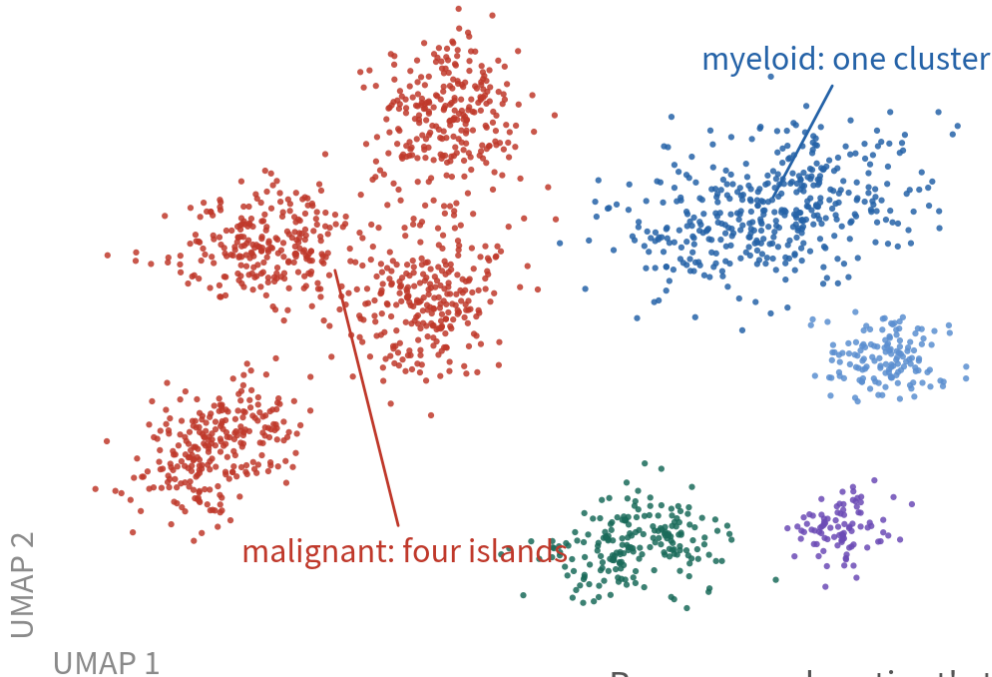
Every tissue has its quirks; these two are where tumour data misleads most

Phenomenon 1: malignant cells cluster by patient, normal cells mix

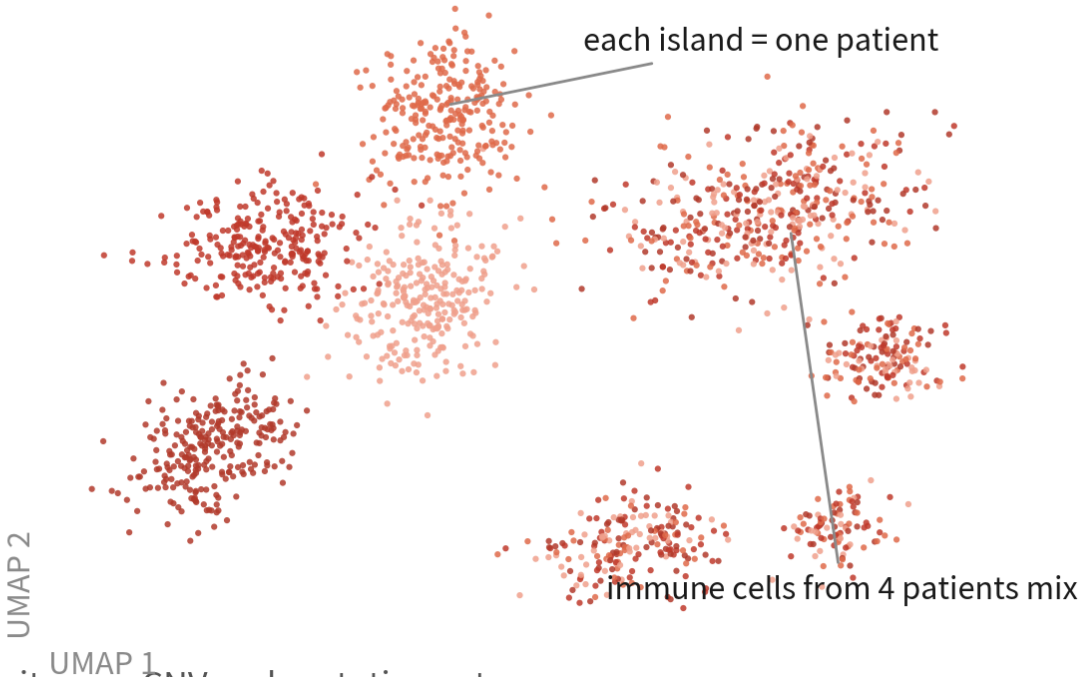
The defining phenomenon of tumour data: malignant cells cluster by patient, normal cells mix across patients

● malignant ● myeloid ● T ● oligo ● vascular

coloured by cell type



coloured by patient

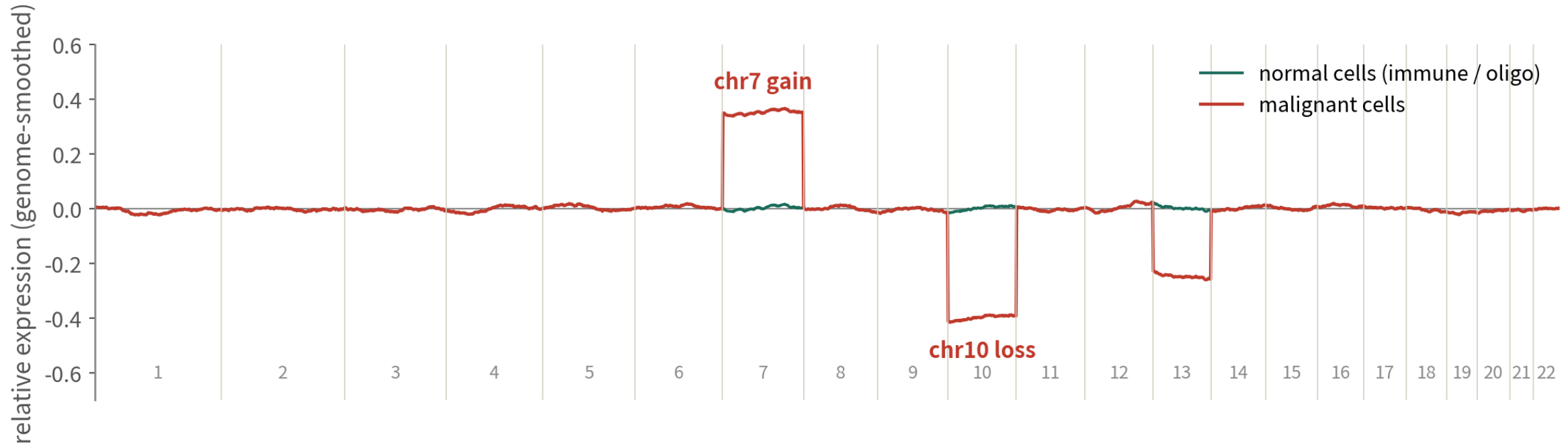


Because each patient's tumour has its own CNV and mutation set.
This is not a batch effect—be careful not to 'correct' it away (Q3)

Each patient's tumour carries its own CNV and mutation set. This is not a batch effect — do not correct it away (Q3)

Phenomenon 2: malignant vs normal is decided by the genome, not markers

A malignant cell's ID is its genome: infer CNV from chromosome-scale expression trends



One gene says nothing; a hundred neighbouring genes on the same chromosome shifting together has only one explanation—the DNA copy number changed. That is inferCNV, and the most reliable malignant-vs-normal evidence in tumour data (hands-on in Q3)

SOX2 and EGFR only say 'glial lineage' — astrocytes express them too. The evidence for malignancy is a whole chromosome arm shifting together: inferCNV

The four continents of the tumour microenvironment

The first cut in annotating tumour data is always these four

immune + vascular + normal

mix across patients; clear markers

- PTPRC: gate for all immune cells
- PECAM1 / VWF: endothelium
- MBP / PLP1: oligodendrocytes — a good inferCNV reference

glial / malignant

islands by patient; markers only

give lineage

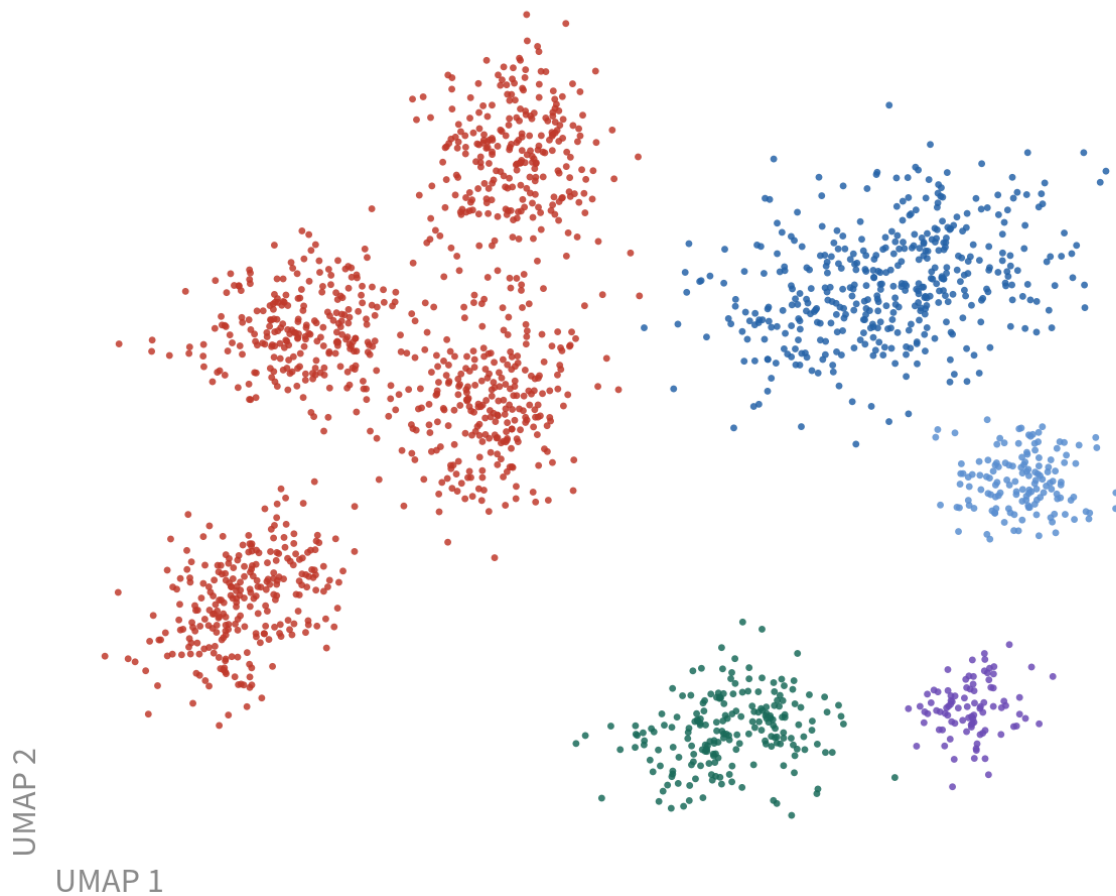
- SOX2 / OLIG2 / PTPRZ1 / EGFR
- malignancy called separately: CNV (Q3)
- then look at states inside malignant cells (MES / AC / OPC / NPC-like)

05

Reading single-cell plots

For most people this is the only skill they will ever need

What a UMAP can and cannot tell you



It can tell you

- ✓ which cells resemble each other (local neighbours are reliable)
- ✓ roughly how many groups, which are connected
- ✓ where a gene lights up (FeaturePlot)

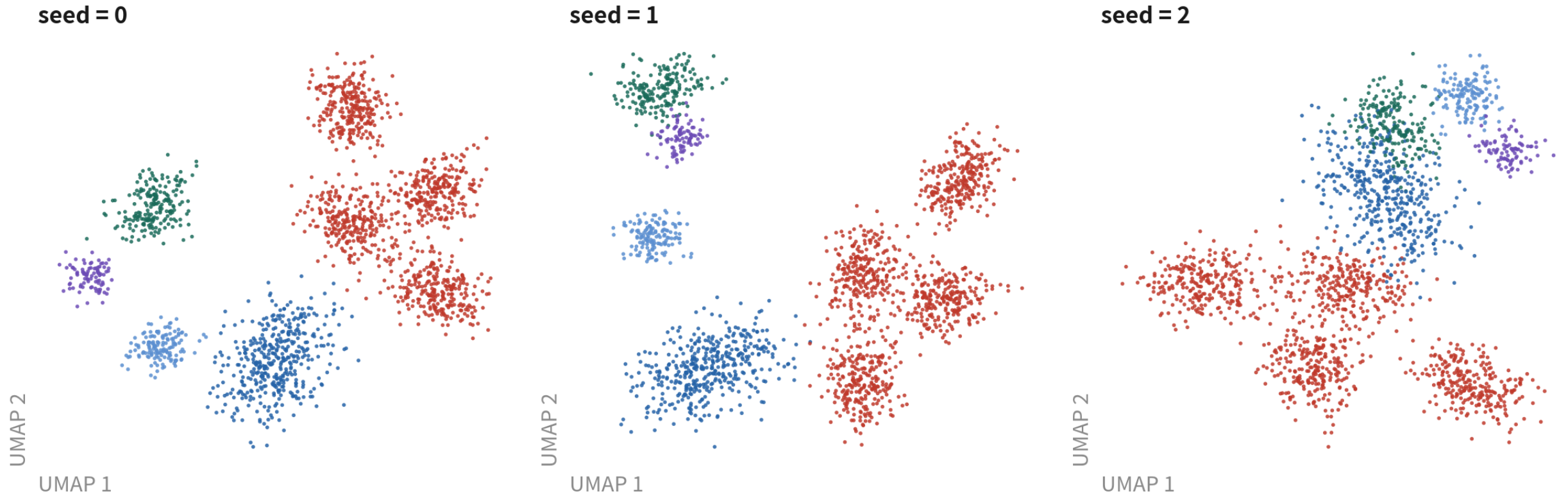
It cannot tell you

- ✗ distance $\neq$ relatedness (a new seed moves it)
- ✗ area $\neq$ cell count (use `table()`)
- ✗ an elongated shape $\neq$ a trajectory
- ✗ four malignant islands $\neq$ four subtypes (those are four patients)

The last line is tumour-specific: four malignant islands are not four subtypes, they are four patients

Same data, only the seed changes

Same tumour data, same code, only the seed changes

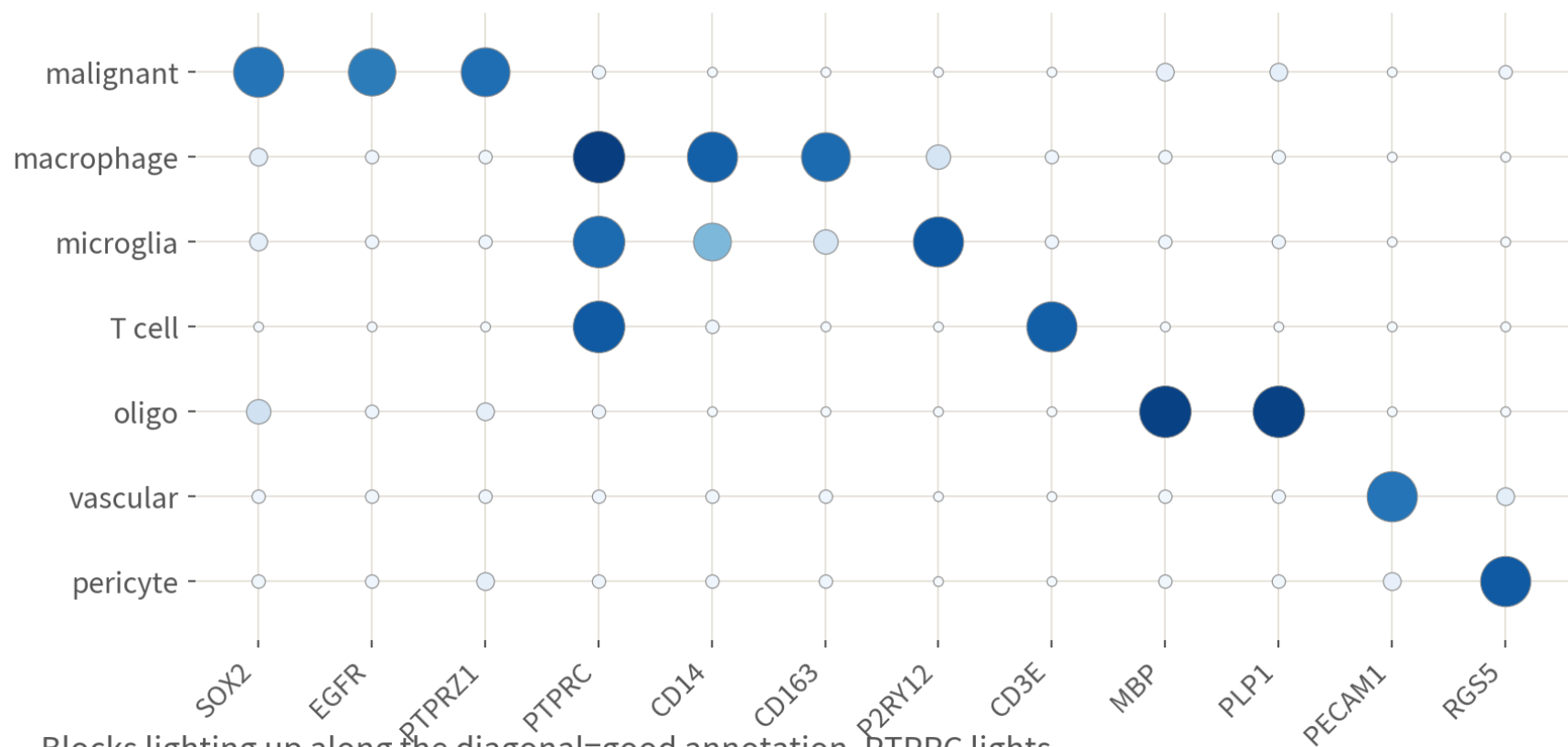


Positions and shapes all change; not one cluster label does — clustering happened in PC space, not on this plot

Positions and shapes all change, not one label does. 'Cluster A sits near B, suggesting relatedness' carries no evidence

Dotplot: read the two dimensions separately

Dotplot: which group expresses what — a TME marker panel



Blocks lighting up along the diagonal=good annotation. PTPRC lights three immune types: it marks the immune continent, not a type

Read the two dimensions separately

- 20%
- 60%
- 100%

size=fraction of cells expressing
colour=expression

Size is the fraction of cells expressing, colour the level. Blocks along the diagonal mean a good annotation

FeaturePlot: colour does not always mean expression

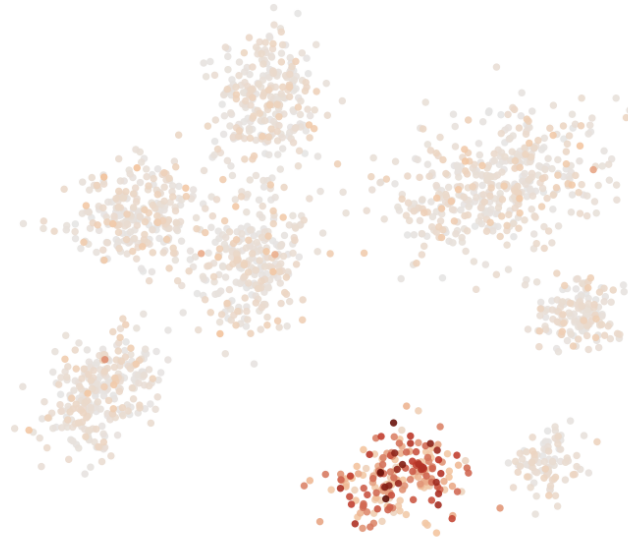
FeaturePlot: 'coloured' can mean three different things

a clean marker: PTPRC



lights only in immune cells,
everything else grey—usable as a
gate

ambient RNA: MBP



bright in oligodendrocytes,
but every cell has a little
—mRNA from lysed cells
drifts into every droplet

a low-detection gene: IL2



a scatter of single bright
cells, no cluster shape—no
conclusion from single cells

A clean gate lights one block; ambient RNA gives every cell a little; a low-detection gene is a sprinkle. First question: does the bright region have the shape of a cluster?

The four most common misreadings

What you see	What you assume	What is true
Two clusters far apart	Biologically distant	UMAP keeps no global distance; a new seed moves them
A big cluster area	Many cells	Area reflects density and parameters; count with <code>table()</code>
A cluster stretched into a band	A trajectory	Could be cell cycle, depth gradient, or a doublet bridge
Malignant cells in four islands	Four malignant subtypes	Four patients; subtypes are read within patients via state scores

Reading Figure 1 of a single-cell paper: five questions first

All five answers live in the methods and supplements; if you cannot find one, discount the conclusion

Question	Why it matters	Where to look
How many patients per group?	n is patients; between-group claims at $n < 3$ are descriptive	Table 1, 'Samples' in methods
How were cells QC'd and doublets removed?	Without doublet removal, 'dual-lineage' types are usually artefacts	'QC' in methods, Fig S1
What was integrated, what was corrected away?	Integrating by patient flattens real malignant differences (Q3)	'Integration' in methods, pre/post UMAPs
How was malignancy called?	'Tumour cells' by markers alone are not credible; look for CNV or mutations	inferCNV / copyKAT heatmap
Is DE done per cell or per patient?	Cell-level tests hand you thousands of false positives (Q3)	'Differential expression': pseudobulk or not

06

Where people get hurt

Three conceptual traps: no error message, just a wrong conclusion

Trap 1: treating cells as samples

What it is

One patient's tumour vs one patient's normal brain, a test over 30,000 cells, 3,000 'disease genes'.



Why it happens

Every cell looks like an independent observation.

What it costs you

Cells from one patient are correlated; n is really 1. Most of those genes will not hold in the next patient.

Trap 2: reporting dissociation stress as tumour biology

What it is

A FOS/JUN/HSPA1A-high malignant subset gets written up as a 'stress-response subclone'.



Why it happens

Clean markers, significant enrichment, and it resembles a published 'MES-like state'.

What it costs you

It is the immediate-early response to enzymatic dissociation, present in every type. Tumour tissue dissociates slowly, so the artefact is strong.

Trap 3: calling malignancy from markers

What it is

'This cluster expresses EGFR and SOX2, so it is tumour.'



Why it happens

Those genes really are textbook GBM markers.

What it costs you

Reactive astrocytes and OPCs express them too. The evidence is CNV; a 'tumour cell' without CNV support will not survive review.

What lies beyond single-cell transcriptomes

This series does RNA only; knowing the rest of the map keeps you oriented when reading papers

Multimodal



RNA + protein / chromatin. CITE-seq adds surface proteins (sharper immune calls); 10x Multiome adds ATAC (regulation, lineage). Same analysis skeleton as this course

Spatial transcriptomics



Visium, Xenium, MERFISH keep position: who sits by vessels, in necrosis, at the core–edge boundary. Single-cell data often serves as the reference for deconvolution

Lineage and genome



TCR / BCR sequencing tracks clones; scDNA-seq or mitochondrial variants track subclones. Tumour-evolution studies pair these with expression

ONE-LINE SUMMARY

**A tumour is an ecosystem and single cell takes it apart;
analysis lives in PC space, the UMAP is a projection;
malignancy is read from the genome, and the patient is the unit.**

Three sentences. Every line of code in Q2 and every decision in Q3 rests on them.

Check yourself 1

Bulk shows CD8A three-fold higher in tumour A than B. The most conservative reading?

- A A's T cells are more cytotoxic
- B A may have more CD8 T cells, or each expresses more — bulk cannot tell
- C A will respond better to immunotherapy

Answer B. Composition change and intrinsic change look identical in bulk. A and C over-interpret; you need single-cell data (and multiple patients) to move in that direction.

Check yourself 2

Four GBM patients: malignant cells form four islands, macrophages one blended cluster. This means?

- A Severe batch effect; integrate
- B Four malignant subtypes
- C Expected: malignant cells carry patient-specific CNV, normal cells do not

Answer C. Macrophages blend well, so technical batch is small; malignant islands come from genomic differences. Forcing integration erases real patient biology.

Check yourself 3

The strongest evidence that a cluster is malignant?

- A High EGFR and SOX2
- B Whole chromosome arms (e.g. chr7 gain, chr10 loss) shifting together
- C Sitting far from immune cells on the UMAP

Answer B. Markers only give lineage and UMAP distance is not evidence; CNV is genome-level evidence and the basis of inferCNV.

NEXT

Q2: Hands-on I — the standard pipeline on one sample

Concepts done, now your fingers. Next episode runs 10x's public GBM 5k (5,604 cells) from loading to annotation, with three practice scripts — every block copy-ready.